



Question 1 (12 marks)

(a) Explain the importance of glycine in the formation of collagen triplex helix. [2max]

- Glycine is small/smallest amino acid; (*Bonus*: for identifying R as Hydrogen)
- Allows formation of **kinked** helix;
- Allows **tight fitting** of the 3 helices/ triple helix/tropocollagen/does not disrupt triple helix;
- By decreasing steric hindrance;
- Allows formation of **interchain hydrogen bonds** are formed;
- between the peptide -NH group of the glycine residues and the peptide -CO groups of residues on the neighbouring chains;

® glycine is hydrophobic and this makes collagen insoluble – as question asks for the formation as in structure.

NB: collagen triple helix = tropocollagen = basic unit/ building blocks of collagen

(b) Describe how association between triple helices allows the formation of fibers with high tensile strength. [3 max]

1. Each tropocollagen interacts with other tropocollagen **running parallel** to it;
 2. **Covalent bonds** formed;
 3. between the side-chains of the **lysine residues** between tropocollagen;
 4. **Staggered arrangement** of tropocollagen in parallel chains;
 - ® ends of tropocollagen is staggered.
 5. No weak areas/ more difficult to break;
 6. Cross-links hold many tropocollagen side by side, forming a collagen fibril;;
 7. Collagen fibrils further aggregated to form collagen fibres;
- Allows the formation of a very stable structure with no capacity to stretch;

(c) Distinguish between gene mutations and RNA splicing mutations. [2]

<i>Gene mutations</i>	<i>RNA splicing mutations</i>
involve changes at the DNA level;	involve changes at the RNA level;
involve insertion, deletion, substitution of base sequences; (mention of any 2)	involve the joining of exons/ removal of introns;
Genetic mutations occur in prokaryotes and eukaryotes;	RNA splicing mutations occurs in eukaryotes only

Gene mutations are inheritable but not RNA splicing mutations. (NB: may not be true as the RNA splicing mutations involved proteins which are coded for by DNA.) (1/2 m for complete comparison)

(d) Explain why gene mutations do not always produce mutated collagen whereas mutations involving RNA splicing will always produce mutated collagen. [3]

1. Gene mutations that involve **substitution**;
2. May result in the **same amino acid** being coded for;
3. **Degenerate** code/ same amino acid being coded for by different codons/ a specific example;
 ® degenerative code

An example of RNA splicing mutation (any one -1 mark each)	Effect (i.e. production of mutated collagen) (any one -1/2 mark each)
different combinations of exons being produced	Different primary sequences of amino acids resulting in different protein(mutated protein)
an exon is lost/ wrong excision of exons	Large number of bases and hence amino acids is lost/ as above;
introns not removed	Introns translated and additional amino acids change the protein structure;

(e) Explain why. [2max]

- Cellulose is made up of **β-glucose/ carbohydrates/ monosaccharides/sugar monomers**;;
- **Not coded for by DNA/** DNA sequences only codes for amino acids/ proteins;
- Any change (in cellulose) is due to enzymes (proteins) involved in the formation of glycosidic bonds (which is coded for by the DNA) and hence effect is not direct;
- Collagen is made up of amino acids;

Question 2 (13 marks)

(a) (i) Calculate the average size of the introns for the albumin gene. (Show your workings.) [1]

- $25 - 2.1 = 22.9\text{kb}$;
- $22.9/14 = 1.6\text{kb}$;

(ii) Analyze the relationship between gene size and the number of introns. [2]

- Generally, the bigger the gene size, the higher the number of introns;
- Ref. to insulin gene (1.7kb) with 2 introns and collagen gene (38kb) with 50 introns;
- The general trend has exceptions;
- Ref. to phenylalanine hydroxylase or dystrophin genes;

(iii) Determine the maximum number of amino acids that could be produced by translating the phenylalanine hydroxylase mRNA. (Show your workings.) [1]

- $2400 - 3 = 2397$; (3 bases for the stop codon)
- $2397/3 = 799$ amino acids;

(b) Describe how a eukaryotic cell turns on its genes in its chromatin fibres. [2]

- Histone acetylation / using histone acetylases;
- To decondense the chromatin;
- Chromatin remodeling / ATP-dependent chromatin remodeling complexes;
- Displace the nucleosomes;
- Facilitates assembly of general transcription factors and RNA polymerase;
- Transcription initiation;

(c) State three differences between transcription and replication. [3]

Features	Transcription	Replication
Template	DNA template strand;	DNA parental strands;
Raw material	Free ribonucleotides;	Free deoxyribonucleotides;
Enzyme	RNA polymerase;	DNA polymerase;
Product(s)	mRNA, rRNA and tRNA;	Two daughter DNA molecules;

(d) With reference to Figure 2.1, describe how the endoplasmic reticulum manufactures and packages secretory glycoproteins. [4]

- Signal recognition particle (SRP) binds to ribosome and emerging polypeptide;
- With signal peptide / signal sequence;
- Arrests translation;
- Binds to receptor on endoplasmic reticulum;
- SRP released;
- Translation continues and pushed through lumen of endoplasmic reticulum;
- Signal peptidase cleaves off signal peptide;
- When protein is in endoplasmic reticulum, proteins are modified;
- Proteins depart in transport vesicles;

Question 3 (14 marks)

(a) Briefly describe the structure of a Lambda (λ) phage. [3]

1. Nucleic acid core/ DNA core; carries information for synthesis of viral proteins/ viral glycoproteins/ viral enzymes.
2. Protein coat/ capsid; encloses the nucleic acid/ nucleocapsid/; made up of capsomeres; protects the viral genetic material from being degraded;
3. A tail; a hollow tube through which the viral genetic material passes during infection/ binds to receptor sites on host cell membrane/ injects viral genetic material into host cell
4. Extra: It is non-enveloped/ naked;
5. **NB:** This bacteriophage does not possess contractile sheath and base plate. Any functions though correct but attributed to these two will be marked wrong.

(b) Suggest an explanation for the flat part of the graph, labelled A, as shown above. [2]

1. Bacteriophage undergoing lysogenic cycle/ not making new viral components;
2. Prophage/ integrated into host genome/;
3. Viral replication is suppressed/ genes for viral assembly suppressed/ no new phages formed;
4. No host cell lysis/ breakdown of bacteria/ *E. coli*;

(c) Account for the rise in PFU shortly after UV irradiation. [3]

1. Bacterial cells carrying the integrated viral DNA/ prophage is activated/ induced/ an induction event;
2. Lytic cycle;
3. Prophage excised from bacterial/ host chromosomes/ DNA/ genome;
4. Bacterial enzymes transcribe and translate the viral DNA into viral proteins/ viral components;
5. Assembly of viral DNA and proteins into infectious bacteriophages/ viral particles;
6. Bacteriophages are released from host cells following lysis of bacteria/ breakdown of bacterial cell wall;

(d) Describe the differences in the life cycle of T4 phage and HIV virus. [4]

T4 (lytic)	HIV (lysogenic)
Attachement of base plate of tail fibre to specific reseptors on surface of cell membrane	Attachment of glycoprotein/ gp 120 to CD4 reseptors
Contraction of contractile sheath to inject viral DNA into host cell cytoplasm	Endocytosis- fusion of viral envelope with host cell membrane
Uncoating is not required as the capsid is left outside of the host cell	Uncoating is required as the nucleocapsid enters the host cell
Expression and replication of viral genome for the assembly of new viral phages using host cell machinery	Reverse transcriptase to synthesise DNA from viral RNA followed by integration into host cell genome by integrase.
Causes host cell lysis due to action of viral lysozyme	Release is through either budding or lysis depending on type of host cell.

(e) Suggest how a viral infection may cause the immune system to turn against its own body cells. [2]

1. Viral glycoproteins integrated into host cell membrane;
2. Changing the antigenic surface of host cell membrane;
3. Host cell being recognised as foreign;
4. By antibodies;

Question 4 (14 marks)

(a) Describe **two** differences between inheritance caused by polygenes and multiple alleles. [2]

	Polygenic	Multiple alleles
Number of genes involved (for the same characteristic)	Several;	One;
Effect of the individual alleles	Additive/quantitative;	Non-additive/ qualitative;
Effect of environment	Most are affected by environment;	Not affected by environment;
Type of variation	Continuous variation/ phenotypic classes ranging from 1 extreme to another	Discontinuous variation/ phenotypes falling into discrete classes without intermediates;

There must be comparison done between relevant points.

(b) Explain how the data for the parents and F1 support the fact that the inheritance of flower length is polygenic. [2max]

- Since both parents are homozygous at their gene loci i.e. genetically identical;
- but there is still a range of values seen/ continuous variation (same for F1 if included, mark awarded once only);
- variation is due to **environment**;
- F1 must be heterozygous at all these gene loci;
- The variation of F1 is intermediate between both parents;
- the alleles act in an **additive** manner;
- QV: 50 – 70 mm;

(c) Account clearly for the difference in variation shown by F1 and F2. [5max]

- Difference: F2 shows a wider variation compared to F1;
- F1 – variation is due to only environment;
- F1 - individuals all genetically identical/ all heterozygous;
- Quote value: accept a range of 40 – 90 mm;
- Genotypes of F2 more varied than F1;
- Formed as a result of **meiosis** in the formation of gametes (from F1);
- Processes such as independent assortment of homologous chromosomes);

- Crossing over (between linked genes forming new linkage groups);
- Random fusion of gametes during fertilization;
- Different genotypes will respond to the same environment differently/ $\neq$;

NB: data in the graph shows phenotypic variation, not genotypic differences.

(d) Assuming that the plant is **diploid**, how would the results for F2 differ if flower length is caused by one gene with two alleles showing codominance relationship? [3 max]

- 3 different phenotypic classes/ flower length; (mark given only if student did not mention about the phenotype of the 3 classes.)
- $\frac{1}{4}$; showing the flower length of the parent with *longer* flower;
- $\frac{1}{4}$; with flower length similar to the parent with *shorter* flower length;
- $\frac{1}{2}$; showing an *intermediate* length between both parents;
- Distinct classes with no intermediates in between these classes;

NB: Students should clearly state the phenotype of the 3 classes and not just mention that there will be 2 classes showing the parental phenotype. They should not use the average length of the long and short flowers since if this is controlled by 1 gene with 2 alleles showing codominance, then there should only be long, short and intermediate length flowers.

(e) When comparing variation in polygenic characters between two groups of organisms, it is often more useful to compare standard deviations rather than ranges. Explain why. [2]

- Range only shows the highest and lowest value/extremes; and hence is influenced by single "outlier"/anomalous data;;
- SD shows dispersion/spread of data about the mean;;
- SD allows statistical use/ test of significance/ test of significant differences between groups;;

Question 5 (13 marks)

(a) (i) State the resting potential for the cardiac action potential shown. [1]

1. Between (-90 mV to -95 mV)

(ii) With reference to the information shown in Table 5.2, suggest how the resting potential can be maintained in the heart muscles using sodium and potassium ions. [3]

1. $[Na^+]$ outside the cell $>$ $[Na^+]$ inside the cell;
2. $[K^+]$ outside the cell $<$ $[K^+]$ inside the cell;
3. Movement against concentration gradient;
4. Requires energy/ ATP;
5. Possible involvement of a protein pump;
6. More Na^+ pumped out than K^+ pumped in;
7. Nett loss of +ve ions from inside the cell thus inside -ve;

(b) Using the information given in the Figure 5.1, briefly explain the changes in Phase 0. [3]

1. Membrane potential increases from -100mv to +50mV;
2. Opening of Na^+ gates upon stimulation;
3. Influx of Na^+ into the cell;
4. Down electrochemical gradient;
5. Nett inflow of +ve ions;
6. Membrane potential becomes less negative/ becomes more positive
7. Membrane is depolarized.

NB: R- membrane potential increases

(c) (i) Describe a difference at Phase 2 between a nerve cell and a cardiac heart cell. [1]

1. The fall in membrane potential in a nerve cell is more drastic/ shows a steeper gradient/ faster rate/ longer time in
OR
Nerve cell only shows Phase 3/ Phase 2 is absent in nerve cells.

(ii) Phase 2 in the cardiac potential is known as the 'plateau' phase and it is caused by the inward flow of Ca^{2+} . Suggest how this may explain for the difference in (i). [2]

1. The inward flow of Ca^{2+} is equal to that of the outward flow of K^{+} (Any references to the inward diffusion of Ca^{2+} and outward diffusion of K^{+});
2. Ca^{2+} permeability declines while the K^{+} permeability increases with time

(iii) Cardiac arrhythmia is term used for any abnormal electrical activity in the heart. As a result the heart may beat too fast. Calcium channel blockers such as Verapamil are often used to treat this condition. Suggest the action of Verapamil in controlling this symptom of arrhythmia. [2]

1. Interrupts inflow of Ca^{2+} during plateau/ Phase ;
2. Prolongs the refractory period/ depresses Phase 2 and 3;
3. Reduces the rate of heart contraction;
4. Heart rate declines.

(d) Draw into the figure the duration of the 'refractory period'. [1]

Question 6 (14 marks)

(a) (i) State which genus does the orang-utan belong? [1]

- *Pongo*;

(ii) State which order does the chimpanzee belong? [1]

- Primates / Primata;

(b) Explain why this classification is both hierarchical and phylogenetic. [2]

- Hierarchical because each level of category names is contained within a more inclusive category;
- Species within genus, genera within family, etc.;
- Phylogenetic because the hierarchy of names indicates the degree of relatedness among organisms;
- Ranking within more inclusive taxa may reflect recency of common ancestry;

(c) Use the information in the diagram, state the present-day species of finch evolved first on the Galapagos islands? Explain your answer. [2]

Finch species:

- *Certhidea olivacea*;;

Explanation:

- Lineage leading to *Certhidea olivacea* diverged first from the original species of finch from South America;
- Other species diverged later from other ancestries;

(d) Explain why evaluating the DNA base sequences can work out the evolutionary relationship between the species of finches. [2]

- All known life based on nucleic acids;
- Some genes / certain regions of genomes appear to evolve at constant rates;
- Extent of changes to DNA base sequences;
- Determine the extent of relatedness;

(e) (i) Suggest how these different species of finch might have evolved from one species. [3]

- Variation in beak shape present in original population;
- Selection of individuals with beaks suitable for eating small seeds on one island/selection of beak suitable for eating large seeds on another;
- Those with the most suited beaks (survived) to reproduce/pass on their alleles/genes for beak shape;
- Allele frequency changed on each island / populations became genetically different;
- No gene flow between islands/geographical isolation;
- So that individuals from different islands unable to reproduce to produce fertile offspring;

(ii) Describe how biogeography may provide support for the evolution of the different species of finch. [3]

- Biogeography is the geographic distribution of species;;
- Species are found where they are because they evolved from ancestors that inhabited those regions;;
- The Galapagos are colonized by finch species from the nearest mainland, South America;
- They adapt to their new environments;

(3 max.)

Section B: Essays

Question 7(a)

- Sequence of bases in gene/on DNA determine amino acid sequence;
- Transcription to form mRNA;
- Codons on mRNA;
- Reference to base-pairing;
- A to U, T to A and C to G;
- Carries the information for the protein to be synthesized;
- Shine-Dalgarno sequence binds to small subunit of ribosome for protein synthesis initiation;
- Specific tRNA for each amino acid;
- Has anticodon to bind to mRNA codon;
- 3' end has an attachment site for amino acid;
- To bring the appropriate amino acid to the mRNA for protein synthesis;
- Complex of rRNAs and ribosomal proteins;
- Form the large and small subunits of ribosomes;
- Role of A and P sites on ribosomes in translation;

Question 7 (b)

Features	Prokaryote	Eukaryote
Control at chromosomal level	- No control at chromosomal level	- Controls at chromosomal level
	- DNA not associated with histones, hence no histone modification possible	- Involved histone acetylation and chromatin remodeling
Control at transcriptional level	- Operator involved in control of transcription; operator is proximal/near to the genes under its control	- Involved control elements; that may be proximal/near (e.g. upstream regulatory elements) or distal/further (e.g. enhancers and silencers)

		Extra: - Further involved DNA methylation that reinforced transcriptional repression
	- Some operons (e.g. lac operon) underwent both positive (catabolite activator protein) and negative regulation (repressor)	- Involved specific transcription factors (e.g. activators and repressors); combinatorial control and coordinated control involved OR - Control elements and specific transcription factors increased (activated) or decreased (repressed) the rate of transcription of the target gene
Type of mRNA produced	- Polycistronic mRNAs	- Monocistronic mRNAs
	- Operon system: multiple related structural genes controlled by the same promoter	- Each gene controlled by its own promoter
Control at post-	- Transcription and translation occur simultaneously due to	- Nuclear membrane separates the processes of transcription

transcriptional level	absence of a nuclear membrane	and translation
	- No post transcriptional modifications occur	- Post transcriptional modifications occur. Eg -Alternative RNA splicing, - Addition of 5' cap, 3' poly(A) tail occurred Any one
Control at translational level	- Translational controls via 1. by obscuring ribosome binding sites 2. by inhibiting exonucleases 3. by inhibiting translation by ribosomes 4. by antisense RNA molecules Any one	- Translational controls via 1.by life span of mRNA molecules 2.by general RNA-degradation machinery 3.by microRNAs (miRNAs) 4.by blocking translation initiation Any one
Control at post-translational level	- Protein modifications, e.g. - Methylation of native DNA allow the bacteria to protect themselves from infection by bacteriophage - Phosphorylation of glycerol kinase in <i>Streptococcus faecalis</i> probably serves to regulate the degradation of glycerol	- Protein modifications, e.g - Cleavage and chemical modifications - Protein degradation

Question 7(c)

Features	Gene Mutation	Chromosome Mutation/ Chromosome Aberration
Definition	Change in the base sequence / nucleotide sequence of the DNA; Occurs at a <u>single locus</u> on a chromosome;	Chromosomal aberration or mutation is defined as a change in the structure of a chromosome; or the number of chromosomes ; <u>Several gene loci</u> are involved;
Mechanism	Brought about by deletion, insertion, substitution or inversion <u>of bases</u> ; ® Answers presented in a table (or in prose) that does not explain that deletion, insertion etc that are stated refers to changes in the bases.	Brought about by: 1. Deletion, inversion, <i>translocation</i> or <i>duplication of the chromosome</i> ; 2. Non-disjunction (of homologous chromosomes or sister chromatids during mitosis or meiosis);
Results	Only 1 gene is affected (and hence 1 polypeptide product is affected);	Effect is usually more profound as several genes are affected;
	Base substitution may lead to the formation of new alleles;	Base sequence of individual gene usually not affected; Affects the: 1. Positions of several genes on a chromosome; 2. The number of genes on a chromosome; 3. Number of chromosomes in a cell; (max of 1 mark for this point) a) Loss or gain of 1 chromosome/ aneuploidy; b) Loss or gain of sets of chromosomes/ polyploidy;
Occurrence	Occur anytime (could be due to mutagens or reading error in DNA replication); (NB: For gene mutation as well as chromosomal aberration in chromosome structure, these are random events and therefore can occur anytime.)	<i>For chromosome aberration in structure:</i> Occur anytime (could be due to mutagens); <i>For chromosome aberration in number:</i> Occurs when homologous chromosomes or sister chromatids fail to separate; During mitosis and/or meiosis;

Frequency	More frequent as there are many more genes than chromosomes; (NB: it is necessary to qualify why the frequency of gene mutations appear to be higher)	Less frequent;
Evolutionary importance	Involves introduction of new alleles into the gene pool; (NB: point awarded for new alleles once only)	Involves reshuffling of existing alleles in the gene pool;
Consequence	<i>(Can have different consequences ranging from no effect, mutated protein produced to possible beneficial mutations.)</i> May have <u>no effect</u> eg. degenerate code i.e. substitution leads to same amino acid being coded for/ in non-coding regions of the DNA; May be <u>drastic</u> as in frame-shift mutations where a mutated polypeptide can be produced/ truncated polypeptide; May have <u>beneficial mutations</u> (for natural selection to work on) opposed to no beneficial chromosomal mutations; (NB: Not all gene mutations are less severe than chromosomal mutations. 1 gene can code for a protein that is required in a pathway that is crucial is providing material for several other pathways and hence the consequence can be fatal.)	Generally will cause the expression of a <u>different phenotype</u> eg. mental retardation as in Down's syndrome

Question 8(a)

Features	Phospholipids	Triglycerides
Components	1 glycerol, 2 fatty acids, 1 phosphate	1 glycerol, 3 fatty acids
Property	Amphiphatic OR Hydrophilic/polar (phosphate) head + hydrophobic/ non-polar (hydrocarbon) tail	Non-polar/ uncharged/ no polar groups OR Hydrophobic/ insoluble
Function	Major component of cell membrane	Storage compound/ storage energy/ insulation/ buoyancy
Structure related to function	Forms a bilayer Allow compartmentalization Formation of organelles Structure of bilayer is stable because polar phosphate groups interact with aqueous media outside of cell as well as cytoplasm	Stored in huge quantity without affecting osmotic potential of cell
	Weak interactions between hydrocarbon tails makes the phospholipids fluid which allows vesicles to form/ fuse	High number of C-H bonds/ higher ratio of H:O yields more energy upon oxidation compared to phospholipids.

Question 8(b)

1 gene (max 3 marks)

- F2 shows a 3 dominant:1 recessive characteristics;
- 1 gene with 2 alleles with complete dominance and recessive relationship;
- F1 shows intermediate/ a characteristic that is in between the 2 parents;

- F2 shows a 1:2:1 ratio;
- 1 gene with 2 alleles with both alleles showing incomplete dominance/ codominance;
- If male and females show a difference in the expression of a characteristic/ more males are affected by a particular condition/reciprocal crosses give different results with one gender being more affected by a particular condition;
- the gene is sex-linked/ X-linked;

2 genes(max 6 marks)

- F2 shows 9:3:3:1 ratio;
- 2 genes each with 2 alleles showing complete dominance/recessive relationship;
- The 2 genes do not interact with each other/ **independently assort**;
- Deviation from 9:3:3:1 with *predominance of parental types*;
- Recombinants in smaller proportion because crossing over is by chance;
- Linked genes/ genes found on the same chromosome;
- Do not assort independently;
- 3:1 ratio means that the 2 genes are totally linked and no crossing over occurs;
- F2 shows fewer than 4 phenotypic classes/ modification of 9:3:3:1 ratio;
- Epistasis which is 2 genes interacting to produce 1 characteristic;
- one gene mask or modify the phenotypic expression of another gene
- 12:3:1- dominant epistasis;
- Where presence of a dominant allele at **1 locus** prevents the expression of a **second gene**;
- 9:3:4 – recessive epistasis;
- Where presence of a *recessive condition* at 1 locus prevents the expression of a second gene;
- 9:7 – complementary gene action;
- a dominant allele must be present in both loci to express the trait;
- 15:1 – duplicate gene action;
- A dominant allele in either loci will give the same phenotype;

Polygenes (2 marks)

- 1 characteristic – showing a graduation from 1 extreme to another;

- Normal distribution;
- Most of the time affected by environment;
- Many genes, each acting additively;

Question 8 (c)

Stages	Insulin	Glucagon
Reception	• Tyrosine kinase receptor;	• G-protein coupled receptor;
Transduction	• Dimerization; • Phosphorylation of tyrosine residues; • Activation of relay proteins;	• Receptor activated and changes shape; • Activates G protein; • Activation of adenylyl cyclase to form cAMP; • Activates kinases
Response	• Facilitates transport of glucose into cells; • Increasing number of glucose carriers; • Synthesis of glycogen from glucose;	• Facilitates release of glucose out of cells; • Breakdown of glycogen by glycogen phosphorylase;